

Feeding

Most rodents have a plant-based diet that may include leaves, fruit, seeds, and roots. However, many species have alternative diets: water rats and the wood mouse eat snails; rice rats take young turtles; muskrats eat clams and crayfish; the southern grasshopper mouse eats ants and scorpions; and the black rat scavenges in human food supplies. To assist digestion, rodents have a large cecum, a blind-ending sac in the large intestine. This contains bacteria that break down cellulose, the main component of plant cell walls, into digestible carbohydrates. In some rodents, after food is processed in the cecum, it is ejected from the anus and is eaten again. Once in the stomach, the carbohydrates (amounting to 80 percent of the energy contained in the food) are absorbed. This highly efficient process, known as refection, leaves only a dry fecal pellet to be excreted.

OMNIVOROUS DIET

Some rodents are omnivorous: they feed on both plants and animals, depending on availability. This spiny mouse eats mainly vegetation but will also feed on insects.



HERBIVOROUS DIET

Most rodents are herbivorous, eating only plants. The European water vole shown here feeds on aquatic and land plants. Food may be stored for consumption during winter shortages.



CAPYBARA FAMILY

Not all rodent species breed as prolifically as mice and rats do. The capybara, for example, usually produces only one litter a year unless conditions are particularly favorable. Litter size varies between one and 8 but is usually 5. Capybara offspring are well developed at birth and are soon able to follow their mother and eat solid food.

SOCIAL ANIMALS

Among rodents, many species live in organized communities, although some are solitary. These black-tailed prairie dogs, like most ground squirrels, are highly sociable. They live in a system of burrows called "towns," each of which may cover an area up to $\frac{2}{5}$ square mile (1 square km). The interconnected burrows in a town provide a refuge from predators and a safe place to rear young. Within the town, prairie dogs form subgroups known as coteries. Members of a coterie act cooperatively; for example, to defend their territory.

Rodents and people

Some rodents, mainly rats and mice, are considered pests by people because they are often in direct competition with humankind (they occupy the same habitats and eat the same food) and are highly adaptable. Rodents consume over 39 million tons (40 million tonnes) of human food every year, contaminate stored food with their urine and fecal pellets, and are known to transmit more than 20 disease-causing organisms. Although some control of rodent populations is brought about by the use of traps and poisons, many species are sufficiently intelligent to learn to avoid such measures. Only a few of the 2,478 species of rodents, however, are genuine pests: many benefit people, for example, by destroying insects and weeds or by maintaining the health of forests by spreading fungi. Beavers and chinchillas are farmed for their fur, while rats, mice, and guinea pigs are kept as pets and are used extensively in medical research.



GNAWING DAMAGE

All rodents use their well-developed incisor teeth to gnaw. A beaver can fell trees, such as this birch, by gnawing through the trunk. Branches and smaller trunks from the tree are used to build a lodge or to dam a river.

